

IE 492 Senior Project Final Report Evaluation Rubric (40% of project grade)

Part A- Written report (20% of report grade)

Performance Criteria	Outcome	Weight	(10,9) Exemplary	(8,7,6) Proficient	(5,4,3) Apprentice	(2,1,0) Novice	Score
Organization	g	0.4	Report is clear, logical, and organized. Easy to follow reasoning.	Report is generally clear and well organized. A few minor points may be changed.	Report is difficult to follow. Many points unclear. Organization is haphazard.	Reader is confused. Most points unclear. Direction and focus of report is unclear.	
Content	g	0.6	Project outline completely covered: Technical information at an appropriate level. Key concepts, terms and assumptions explained clearly. Analysis of topic clearly evident. Conclusions stated clearly and supported. References complete and cited properly in the text.	Project outline nearly completely covered: Technical information at an appropriate level. Some concepts, terms or assumptions not completely clarified. Analysis of topic generally evident. Some details left missing in some sections. References acceptable.	Project outline mostly covered: Technical information at a marginal level, many concepts, terms and assumptions unclear or not discussed. Reader gains little information or insight. Several missing points in some sections. Some missing references / some references not cited in the text.	Some major sections missing or does not contain the relevant information. Technical information unacceptable, most concepts, terms and assumptions unclear or not discussed. Reader gains nearly no information or insight. Conclusions vague or not given. Most references missing or not cited in the text.	
Written Delivery	g	0.6	Very good, concise, clear, effective use of language. Document follows the format and it is neat and professional; visual appearance adds value to communication.	Good, some writing errors evident that neither hinders clarity nor distracts reader. Document follows the format and it is neat and professional; visual appearance does not detract from communication. Minor re-writing might improve certain sections.	Significant writing errors present that either hinders clarity or distracts reader. Document follows the format to some extent. Appearance is acceptable, but requires re-writing in most sections.	Unacceptable. Writing errors obscure clarity. Appearance is unacceptable. Format is poor.	
Use of visual aids	g	0.4	Tables, graphs and figures are easy to read and follow, referenced appropriately in the text, and enhance communication. Concepts made clearer, most information easy to see and follow, details minimized and main points stand out	Tables, graphs and figures are understandable, referenced appropriately in the text, and adequately convey information. Most concepts made clear, most information generally easy to read and follow, main points stand out, a few details difficult to follow.	Tables, graphs and figures are acceptable, but difficult to interpret, poorly referenced, or incompletely convey information.	Tables, graphs and figures are inappropriately used or missing, not referenced, or fail to convey information.	

Part B- Design process (80% of report grade)

Student Outcome (a) An ability to apply knowledge of mathematics, science, and engineering

Student Outcome (e) Ability to identify, model, formulate and solve industrial engineering problems

Student Outcome (b) An ability to design and conduct experiments, as well as to analyze and interpret data

Performance Criterion	Outcome	Weight	(10,9) Exemplary	(8,7,6) Proficient	(5,4,3) Apprentice	(2,1,0) Novice	Score
Problem Framing	a,e,c	1	Identifies correctly the objectives, requirements, limitations as well as the (outer) context of the given problem.	Identifies the objectives and limitations of the given problem with minor deficiencies without changing the nature of the problem. Context could have been better described.	Identifies the objectives and limitations of the given problem with some deficiencies with important consequences on the final result. Context is not very clear.	Fails to understand the main components of the objectives, requirements and limitations of the given problem. Context undefined.	
Data analysis and interpretation	b	0.5	Data is processed appropriately and correctly.	Data is processed mostly in an appropriate manner with a minor error	Data is processed mostly in an appropriate manner with a set of minor errors/omissions.	Data needs not addressed sufficiently.	
Problem Analysis and Method Selection	a,e	1	Clearly stated, reasonable assumptions that do not damage the validity of the results. Relevant methodologies surveyed properly and chosen method(s) is appropriate with respect to the nature of the problem.	Assumptions clearly stated. A few major simplifying assumptions that do not seriously damage the validity of the results. Sufficient survey of relevant methodologies Chosen method is appropriate for the problem.	Partial survey of relevant methodologies. Chosen method is appropriate with respect to the nature of the problem with major simplifying assumptions making the validity of the results questionable.	Assumptions unclear/ or may create invalid results. Insufficient survey for possible approaches. Chosen method is not so appropriate with respect to the nature of the problem.	
Correct application of the selected methods	a,e	1	Applies correct industrial engineering / mathematical concepts to formulate models/design alternatives with no errors affecting the problem solution. Solves problem using logical and efficient procedure.	Applies correct industrial engineering / mathematical concepts to formulate models/design alternatives with no conceptual errors. Solves problem using logical procedure with only one or two minor procedural errors.	Applies correct industrial engineering / mathematical concepts to formulate models/design alternatives; Solution procedure involves procedural errors/shortcomings resulting in unsatisfactory, but valid solutions.	Applies incorrect industrial engineering / mathematical concepts to formulate models/design alternatives; or solution contains conceptual or procedural errors resulting in infeasible and invalid solutions..	
Generating and analyzing alternative solutions	c	1	Meaningfully differentiated set of alternatives are considered and why other alternatives are not searched is made clear.	A reasonable set of alternatives considered; other alternatives could have been developed and analyzed.	A limited set of alternatives considered, alternatives are not meaningfully differentiated; other reasonable alternatives were available and should have been considered.	Only one solution is presented and evaluated; better solutions were available and should have been considered.	

Part B continued- Design process (80% of report grade)

Student Outcome (b) An ability to design and conduct experiments, as well as to analyze and interpret data

Student Outcome (c) An ability to design diverse systems including manufacturing, service, logistics, financial and information, to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability

Student Outcome (k) An ability to use the techniques, skills, and modern engineering tools necessary for industrial engineering practice.

Performance Criterion	Outcome	Weight	(10,9) Exemplary	(8,7,6) Proficient	(5,4,3) Apprentice	(2,1,0) Novice	Score
Decision making	b	1	Best solution is recommended based on predefined criteria that adequately address the problem. Derived conclusions are consistent with experimental results obtained by an appropriate experimental procedure, or a sound evaluation procedure.	Evaluation criteria or experimental procedure or evaluation procedure contains some minor errors or partially address the problem. Still, the recommended solution is feasible and valid.	Evaluation criteria or experimental procedure or evaluation procedure contains some errors that may affect derived conclusions.	Derived conclusions are not justified or they are inconsistent with evaluation process. A sound evaluation process or experimental procedure is not implemented.	
Implementation issues	c	1	The issues related with the implementation of the recommended solution is fully discussed, described, assessed wrt the predefined context of the problem.	Some effort for discussing and assessing issues related with implementation, some minor points incomplete or missing.	Discussion and assessment of the implementation issues are only superficial.	No discussions, or assessment regarding the applicability of the recommended solution.	
Integration of techniques, skills, tools necessary for IE practice	k	1	Fully explored all available resources/tools to complete the project, integrated a number of skills, demonstrated advanced use of them including software tools above basic knowledge obtained in class.	Explored resources/tools to find a good solution to the project, some resources were left unexplored; demonstrated acceptable use of them including software tools.	Explored resources/tools until a less than optimal solution is found; application of them including software tools may have errors; opportunities of using other skills are overlooked	Only the approaches suggested by the supervisor considered; going beyond what is presented in class is avoided; use of software avoided or used with difficulty when required.	
Considering non-technical issues (i.e. environmental, societal, ethical, health, safety, sustainability)	c	0.5	All significant non-technical issues identified and clearly considered in the design process.	Most significant non-technical issues considered and incorporated into the design process.	Some non-technical issues considered, some issues not considered or not clearly incorporated into the design process	Non-technical issues either not considered or just mentioned without being considered in the design process.	